The Mole Concept

Chemistry practice worksheet

Study snapshot

Practise the links between mass, molar mass, amount of substance, and particles.

Key ideas

- Amount of substance is measured in moles (mol).
- Use n = m / M to convert mass to amount.
- One mole contains approximately 6.022 x 10^23 entities.
- Keep units visible through every conversion.
- Balance a chemical equation before using mole ratios.

Try the questions on page 2 before checking your work.

The Mole Concept

Quick practice

1. Find the amount in 18 g of water (M = 18 g/mol).
2. How many moles are in 44 g of carbon dioxide (M = 44 g/mol)?
3. How many molecules are in 2 mol of a substance?
4. For N2 + 3H2 -> 2NH3, how many moles of H2 react with 2 mol N2?

Answer notes

1) 1 mol
2) 1 mol
3) About 1.204 x 10^24 molecules
4) 6 mol H2

Review the method, then retry any question you missed.